

Exam Date & Time: 25-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024 COMPUTER VISION [CSE-4031]

Marks: 50

Duration: 180 mins.

Part A

Answer all the questions.

- 1) What is the goal of computer vision? How image can be processed by applying different steps of computer vision? Explain with a diagram (4)
 - A)
 - B) Define smoothing spatial filters. Apply the mean filter on the pixels (1,1), (3,4) and (5,3) to enhance the quality of image. Assume zero padding for borders. (3)

1	4	0	1	3
2	2	4	2	2
1	0	1	0	1
1	2	1	0	2
2	5	3	1	2
 - C) With a necessary graph, explain the Power-law and Image Log transformations used in image enhancement. What is the necessity of such image enhancements? (3)
- 2) Why canny edge detection is superior to other edge detection algorithms? Explain with an example the purpose of non-maximum suppression and double thresholding in canny edge detection. (5)
 - A)
 - B) Consider an image search engine that takes the input image and identifies matching images in the database. The images in the database may be scaled/rotated. Develop an algorithm to extract scale/rotation invariant feature descriptors from the image so that image search engine can apply this algorithm to find two matching images. (5)
- 3) Explain the following geometric transformations on the image with appropriate mathematical representation. (5)
 - i) Rotation ii) Scaling iii) Translation iv) Skewing
 - A)
 - B) Illustrate following camera models with a neat diagram and describe how the scene image is projected onto a image plane using these models. (5)
 - i) Thin lens camera model ii) Pinhole camera model

- 4) What is the objective of object tracking in computer vision? Explain the role of object tracking in motion estimation in videos. (3)
- A)
- B) In a military surveillance video, there is a need to track aircraft across different frames in a video. Develop an algorithm based on optical flow to estimate the motion of the aircraft. (4)
- C) Differentiate between pixel-based motion estimation and feature-based motion estimation based on any three factors. (3)
- 5) How machine learning is different from traditional programming? Explain the working principle of artificial neural networks in object recognition. (5)
- A)
- B) Explain the technique of computer vision that can be applied to solve the following challenges in face recognition. (5)
- i) Noisy image ii) Partial visibility

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